To the Editors-in-Chief
Autonomous Agents and Multi-Agent Systems

Dear Editors,

We submit for consideration as a regular research paper the manuscript *Safe designs, unsafe ecosystems: deployment-layer selection in an evolutionary AI race.*

The paper studies an ecosystem of deployed autonomous agents in which the payoff an agent earns in its interactions is not the quantity that decides whether its design is replicated. Replication is decided by a principal who does not play the game and who internalises only a fraction of the harm the design causes. We call the difference between the two payoff functionals the selection wedge, embed it in a four-design model of an artificial intelligence race, and analyse the resulting replicator and finite-population dynamics.

The wedge reduces to a single control parameter, an effective liability, in which every invasion condition is affine, so all thresholds are closed form. Four consequences follow that are invisible when the two payoffs are identified. Screening designs one at a time before deployment cannot move the long-run outcome, because the design such a filter removes is weakly dominated. The liability that protects a safe population is decided by a composition that selection leaves neutral, and varies by a factor of 77.7 along it, so a population can lose its protection while nothing observable about its behaviour changes. Long-run harm is not monotone in the strength of the accountability instrument: over an intermediate window the instrument taxes the retaliation that conditional safety depends on, and crossing the lower edge of that window lifts unsafe behaviour from none to about a third of rounds. And coupling enforcement to irreversibly diffusing capability makes the loss of a safe regime hysteretic, by a factor that is exactly computable. Together these say that governance of an agent ecosystem binds at the layer where designs are selected rather than at the layer where they act, which we believe is of direct interest to the readership of this journal.

The work is original, has not been published in any form, and is not under consideration elsewhere. No part of it has appeared in a conference proceedings. All code, data and figures are public under the MIT licence at `https://github.com/trungkiet2005/deployment-layer-selection`, and every numerical value quoted in the manuscript is regenerated by that repository and pinned by its test suite. The required information sheet accompanies this submission.

We confirm that all authors have approved the submission and that we have no competing interests to declare.

Yours sincerely,

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